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Impedance spectroscopy and AC conductivity mechanism in GeSn thin films

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Serhiy Kondratenko^{1,*} , Oleksandr Datsenko¹ , Fernando M de Oliveira², Serhii Kryvyi², Kateryna Yablochkova¹ , Petro M Lytvyn⁴, Morgan E Ware^{2,5}, Serhii Maliuta³ , Shui-Qing (Fisher) Yu⁵, Yuriy I Mazur^{2,*} and Gregory J Salamo²

¹ Taras Shevchenko National University of Kyiv, 64 Volodymyrs'ka St., 01601 Kyiv, Ukraine

² Institute of Nano Science and Engineering, University of Arkansas, Fayetteville, AR 72701, United States of America

³ National Technical University of Ukraine 'Igor Sikorsky KPI', 37 Beresteyskyi av., 03056 Kyiv, Ukraine

⁴ V E Lashkaryov Institute of Semiconductors Physics, NAS of Ukraine, 41 pr. Nauki, 03028 Kyiv, Ukraine

⁵ Department of Electrical Engineering, University of Arkansas, Fayetteville, AR 72701, United States of America

* Authors to whom any correspondence should be addressed.

E-mail: kondratenko@knu.ua and yumazur@uark.edu

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Abstract

In this work, we present a study of the temperature- and frequency-dependent AC conductance in CVD-grown GeSn on Ge/Si substrates, complemented by nanoscale Kelvin probe force microscopy and scanning capacitance microscopy. The AC conductivity in GeSn films follows Jonscher's universal power law, $\sigma(\omega) \sim \omega^s$, with $0.15 < s < 0.50$ in the low-temperature region (85–130 K). The weak temperature dependence of s below ~ 110 K suggests the dominance of quantum mechanical tunneling, while its increase with temperature reflects the influence of other mechanisms, such as variable range hopping and/or phonon-assisted tunneling on the overall AC conductivity. We identify the dominant hopping-based AC transport mechanisms and correlate them with film morphology, localized defect states, and spatial variations in carrier concentration and surface potential. These results reveal the decisive role of disorder-induced localized states in low-temperature transport, providing critical insights for optimizing their performance in advanced optoelectronic devices.

1. Introduction

Recently, there has been significant interest in germanium-tin (GeSn) film fabrication due to its tunable band gap, indirect-to-direct crossover, and high carrier mobility, making it a promising candidate for CMOS-compatible optoelectronic and photonic applications [1, 2]. Among these features, the potential to create a direct bandgap transition in group IV material is of particular interest, as it supports the integration of efficient light emitters, photodetectors, and lasers on silicon. It is generally accepted that a Sn content above $\sim 6\%$ – 10% is required to induce the indirect-to-direct transition. However, the high compressive strain arising from the lattice mismatch between GeSn and Ge/Si or Si substrates further increases the required Sn content, thus requiring a high Sn content in materials suitable for practical applications.

Despite the technological promise of high-Sn-content alloys, there are specific difficulties in growing GeSn alloys with a high tin content due to the difference in atomic radii between Sn and Ge (about 13%) and the low solubility of only about 1% of Sn in Ge [3]. In this regard, considerable efforts have been made to develop advanced growth techniques that enable the incorporation of high Sn concentrations while maintaining crystalline quality and minimizing defects. Recent advances in GeSn synthesis techniques, including molecular-beam epitaxy (MBE) [4], chemical vapor deposition (CVD) [5, 6], and high-temperature sputtering epitaxy [7], have enabled the application of GeSn alloys in various electronic and optoelectronic devices. Among these methods, CVD offers a scalable, industry-compatible

method for fabricating GeSn films, though these films exhibit greater disorder than those produced by MBE technique [8]. Nevertheless, CVD-grown GeSn films achieve quality sufficient for critical mid-infrared photonic applications, including photodetectors, light-emitting diodes, and lasers [9].

However, the performance and reliability of these devices can be affected by strain, released above a critical epitaxial layer thickness, leading to the generation of extended defects, such as dislocations, stacking faults, twins, Sn segregation, and Sn precipitation during growth [10, 11]. These defects introduce localized states within the bandgap, acting as recombination centers or scattering sites that reduce carrier mobility and degrade electrical conductivity, thereby limiting the performance of optical and electronic devices. In particular, defects generate an electrical charge background that can either compensate for or increase the effects of doping in the device design, thereby altering the active carrier concentration. Unfortunately, little is known about the background doping of GeSn/Ge heterostructures in the literature [12, 13]. Undoped GeSn alloys have generally demonstrated p-type conductivity due to the introduction of hole-generating defects, such as vacancies and dislocations, with a concentration of 10^{17} cm^{-3} [13–15]. Such unintentional conductivity has been attributed to acceptor-like defect states introduced during strain relaxation, such as a multivacancy complex with an energy level of 10–20 meV above the valence band [16]. Deep-level transient spectroscopy (DLTS) studies report Sn-vacancy (V) complexes in electron-irradiated Ge doped with Sn and phosphorus (P) at 0.19 eV above the valence band top ($E_V + 190 \text{ meV}$) [17], while hole traps at 20–30 meV have been observed in GeSn/Ge heterostructures with 2.3%–3.2% Sn [13]. Further studies have utilized temperature-dependent photoconductivity transient measurements to identify acceptor-like defects at $E_V + (85\text{--}90) \text{ meV}$ attributed to misfit dislocations at the GeSn/Ge interfaces [18].

The influence of such defects on the conduction mechanisms in GeSn alloys under alternating current (AC) conditions remains largely unexplored [8], although frequency-dependent conductance and capacitance measurements are well-established non-destructive techniques for investigating semiconductor thin films, heterojunctions, quantum wells, and electronic devices [19]. At low temperatures, in the presence of a band of disorder-induced localized states, charge transport may predominantly occur through hopping rather than via thermally activated band conduction [20, 21]. Depending on the density of localized states and Coulomb interactions, transport may occur via different hopping mechanisms including Mott variable-range hopping (VRH), Efros–Shklovskii VRH mechanism, and nearest-neighbor hopping, each characterized by distinct temperature- and frequency-dependent signatures. To the best of our knowledge, very few experiments have observed it in GeSn alloys [22, 23].

In this work, we present the first comprehensive investigation of temperature- and frequency-dependent AC conductance, together with Kelvin probe force microscopy (KPFM) and scanning capacitance microscopy (SCM) analysis at the nanoscale, in CVD-grown GeSn thin films. We identify the dominant AC transport mechanisms, analyze the hopping processes, and correlate the results with material morphology and spatial variations in carrier concentration and surface potential arising from disorder-induced localized states. Such insights are crucial for optimizing performance in optoelectronic applications, where light-matter interactions play a pivotal role.

2. Experiments

The GeSn films studied were grown on Ge/Si substrates using an industry-standard ASM EpsilonR 2000 Plus reduced-pressure CVD system, employing SnCl_4 and GeH_4 as precursors for Sn and Ge, respectively, with H_2 as the carrier gas. Before growth, the silicon wafers were cleaned in vapor-phase HF and subsequently annealed at $\sim 1000^\circ\text{C}$ in H_2 to remove surface oxides. The CVD growth procedure, including the $\text{SnCl}_4/\text{GeH}_4$ flow ratio, temperature conditions, and composition/strain control, has been described in detail previously [24–26]. A Ge buffer layer approximately 700 nm thick was first grown using a two-step method on a boron-doped Si (100) substrate with an acceptor concentration of about 10^{15} cm^{-3} , and this sample was used as a reference. First, a $\sim 150 \text{ nm}$ seed layer was deposited at $< 400^\circ\text{C}$ in H_2 carrier gas at a GeH_4 partial pressure of 0.2 Torr. In a second step, the temperature was then increased to 600°C , and after stabilization, the remaining $\sim 500 \text{ nm}$ was grown. A final *in-situ* post-growth anneal was performed at $> 800^\circ\text{C}$. Then, using the same Ge/Si heterostructure as a template, two separate GeSn films were grown under identical conditions with thicknesses of $\sim 200 \text{ nm}$ (sample S200) and $\sim 1000 \text{ nm}$ (sample S1000), each with a target average Sn content of around 12% and different strain. As a result, three samples were selected for further investigation of the effects of strain relaxation and microstructural disorder on carrier transport mechanisms in GeSn films.

Rectangular $4 \times 1 \text{ mm}$ ohmic electrodes were created on the GeSn layer at 170°C for conductivity measurements using Gallium–Indium eutectic Ga75.5%/In 24.5%, $\geq 99.99\%$ (Sigma-Aldrich). The electrode spacing on the sample surface was 5 mm, minimizing the impact of contact potential difference

on transport measurements. The experimental I - V curves were found to be linear in the range from 80 to 300 K at an applied DC voltage of $V < 1$ V, demonstrating ohmic behavior. Admittance spectroscopy measurements were performed over the 80–300 K temperature range using an LCR Measuring Bridge ST2829C (SourceTronic) with a modulation voltage not exceeding 10 mV and a frequency range of 1 kHz–1 MHz. The DC bias voltage was equal to 100 mV to avoid sample heating.

Continuous-wave (cw) photoluminescence (PL) measurements were carried out at 80 K using the frequency-doubled 532 nm output of a Nd:YAG laser as the excitation source. The laser beam was focused to a spot diameter of ~ 50 μm on the sample surface, with an excitation power of 40 mW. The PL emission was collected, spectrally dispersed by a monochromator, and detected using a PbS photodiode.

Raman spectroscopy measurements were performed using a Horiba LabRAM spectrometer equipped with a Si CCD detector and a He-Ne laser at 632.8 nm as the excitation source. An Olympus 100x objective lens was used to create a probing area of ~ 2 μm in diameter, with a circular laser spot on the sample, at a power of ~ 1 mW. Each measurement is the repeated result of five sampled areas, whose data were collected in two acquisition cycles using a 1800 l mm^{-1} grating.

X-ray diffraction (XRD) measurements were performed using a Panalytical X'Pert MRD diffractometer (Almelo, Netherlands) equipped with $\text{Cu}_{K\alpha 1}$ radiation ($\lambda = 0.15406$ nm). The incident beam was conditioned by a four-bounce Ge(220) monochromator and a multilayer focusing mirror, while the diffracted signal was collected using a Pixel detector.

Photocurrent (PC) spectra were excited by the output of a 150 W halogen lamp with a quartz bulb chopped at 120 Hz and dispersed using a prism monochromator. The DC bias voltage of 100 mV was applied between InGa electrodes. The modulated signal was detected with a Stanford Research Systems SR860 lock-in amplifier with an SR570 low-noise current preamplifier. All the presented spectra were obtained at 80 K and have not been normalized to the source flux.

The nanomorphological and nanoelectrical properties of the films were investigated using a NanoScope IIIa Dimension 3000™ scanning probe microscope. SCM measurements were conducted in contact mode using solid platinum tips, featuring a spring constant of 0.3 N m^{-1} and a tip radius of less than 20 nm. Integral amplitude and phase SCM measurements were performed at an AC voltage of 1.5 V and a frequency of 100 kHz.

Local surface potential mapping was carried out using frequency-modulated KPFM (FM-KPFM). Platinum-coated silicon probes with a nominal tip radius of less than 25 nm were used. The measurements were performed in a two-pass (lift-mode) configuration with a lift height of 20 nm, which minimized contributions from surface topography and van der Waals interactions. The tip work function was calibrated using a certified highly ordered pyrolytic graphite ZYB grade standard. All measurements were carried out under ambient conditions, with a temperature of 23 °C and a relative humidity of 40%.

3. Results and discussion

3.1. Structural characteristics

Raman spectroscopy and XRD were used to characterize the structure of the GeSn films. Figures 1(a) and (b) displays the ω - 2θ scans of the samples S200 and S1000 measured across the symmetrical 004 reflections of Si, Ge, and GeSn. The peaks of the Ge buffer layer are observed at $\omega = 33.0394$ deg, the same for both samples, thus meaning in-plane tensile strain in the Ge buffer layer. A simulation showed that the GeSn film in sample S200 consists of two partly relaxed GeSn layers with Sn contents of 12% at the bottom and 14.7% at the top. The Sn content about 12% was found in sample S1000. Additionally, the shape of the GeSn peak indicates a graded GeSn layer with a concave profile of Sn content, with lower and higher Sn concentrations of 7% and 10%, respectively.

The 004 ω - 2θ scan alone is sensitive only to the vertical parameter $a_{\perp} = 2\lambda/\sin\omega$ of the lattice and gives no information about the Sn content when the strain state is unknown. The lateral lattice parameter $a_{\parallel}$ was estimated from the reciprocal space maps (RSMs) in figures 1(c) and (d).

In turn, the vertical and lateral lattice parameters allow for evaluating the relaxed lattice parameter as

$$a_0 = \frac{a_{\perp} + 2\frac{C_{12}}{C_{11}}a_{\parallel}}{1 - 2\frac{C_{12}}{C_{11}}}, \quad (1)$$

where C_{12} and C_{11} are the elastic constants. The Sn content (x) of the GeSn alloy may be determined considering the lattice parameters of Sn and Ge as $a_0 = a_{\text{Sn}}x + (1 - x)a_{\text{Ge}}$. The data for all samples are collected in table 1.

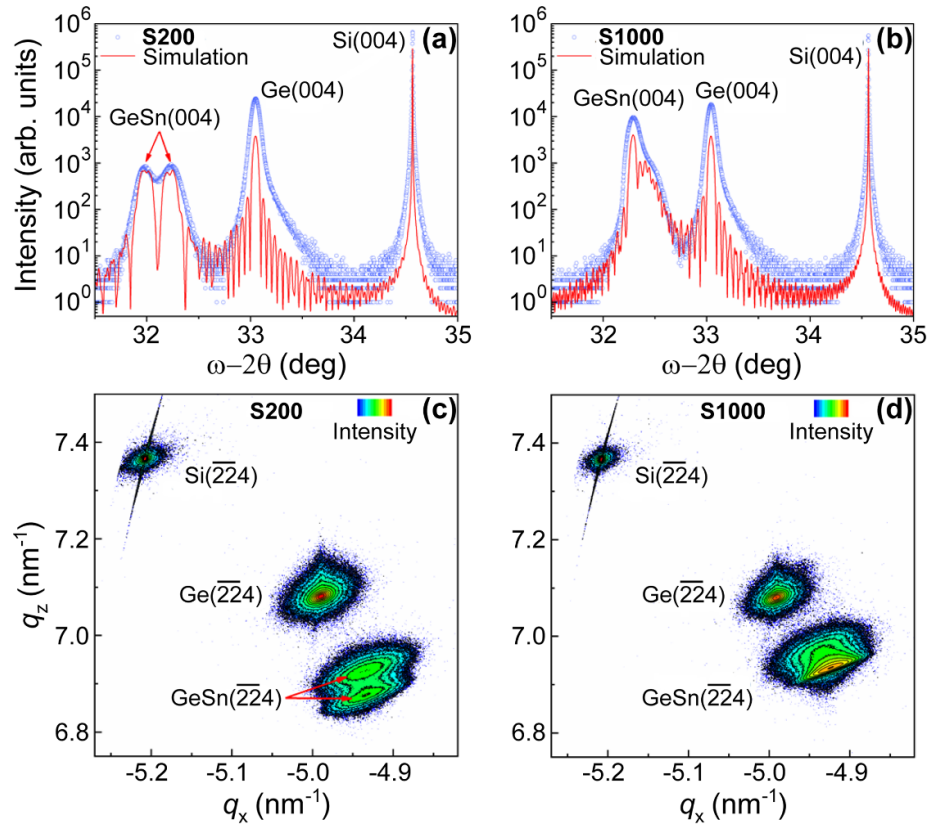


Figure 1. ω - 2θ XRD scans of the samples S200 and S1000 obtained across the symmetrical 004 reflections of Si, Ge, and GeSn (a), (b) and RSM maps around the asymmetrical $(-2-24)$ reflection of GeSn films (c, d).

Table 1. The Sn content and strain in the studied samples derived from XRD and Raman measurements.

Sample	Layer	XRD: Sn, %	XRD: ε_{xx} , %	XRD: ε_{zz} , %	Raman: Sn, %
S200	Ge	—	0.20 ± 0.01	-0.15 ± 0.01	$13.8\% \pm 0.2\%$
	GeSn (bottom)	12 ± 0.5	-0.44 ± 0.01	0.32 ± 0.01	
	GeSn (top)	14.7 ± 0.5	-0.82 ± 0.01	0.64 ± 0.01	
	Ge	—	0.17 ± 0.01	-0.14 ± 0.01	
S1000	GeSn gradient	$7-10 \pm 0.5$	—	—	13.7 ± 0.2
	GeSn (top)	12 ± 0.5	-0.22 ± 0.01	0.19 ± 0.01	

The Raman Ge–Ge mode in samples S200 and S1000 (figure 2) reveals a redshift relative to the reference Ge/Si sample due to tensile strain induced by the GeSn film. Furthermore, the asymmetry of the band indicates the presence of the disorder-activated mode at $\sim 280 \text{ cm}^{-1}$ in the Sn-contained samples. Finally, a peak at $\sim 260 \text{ cm}^{-1}$ attributed to Ge–Sn vibration is notable. The Raman data were used to estimate the composition of the GeSn within the 20 nm-thick top layer, limited by the penetration depth of the 632.8 nm laser line. This was achieved by taking into account the correlation between the strained GeSn lattice constant determined by XRD and the shift of the Raman Ge–Ge mode for each sample [27]. From sampling five areas of each sample, the S200 sample was found to exhibit an average Ge phonon frequency of $290.9 \pm 0.1 \text{ cm}^{-1}$, corresponding to an average Sn content of $13.8 \pm 0.2\%$ near the surface. The average Ge phonon frequency of $289.9 \pm 0.2 \text{ cm}^{-1}$ in sample S1000 corresponds to a similar near-surface Sn content of $13.7 \pm 0.2\%$.

3.2. Photoconductivity and photoluminescence spectroscopy

Figure 3(a) presents the lateral PC spectra of the GeSn/Ge/Si heterostructures and the Ge/Si reference sample measured at 80 K over an energy range from 0.4 to 1.8 eV and normalized to the amplitude of the Si response. In the GeSn/Ge/Si samples, the spectra include contributions from light absorption in the GeSn films, the Ge buffer layer, and the Si substrate. The contribution from the Ge buffer layer in the reference Ge/Si sample is observed in the energy range below 1.17 eV, corresponding to the Si

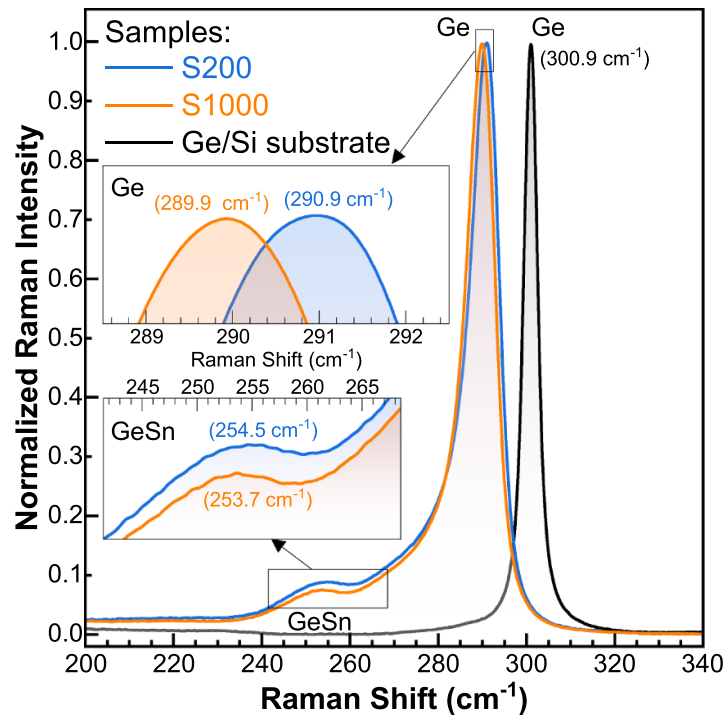


Figure 2. Raman spectra for the GeSn/Ge/Si samples and the reference Ge/Si buffer layer on the silicon substrate.

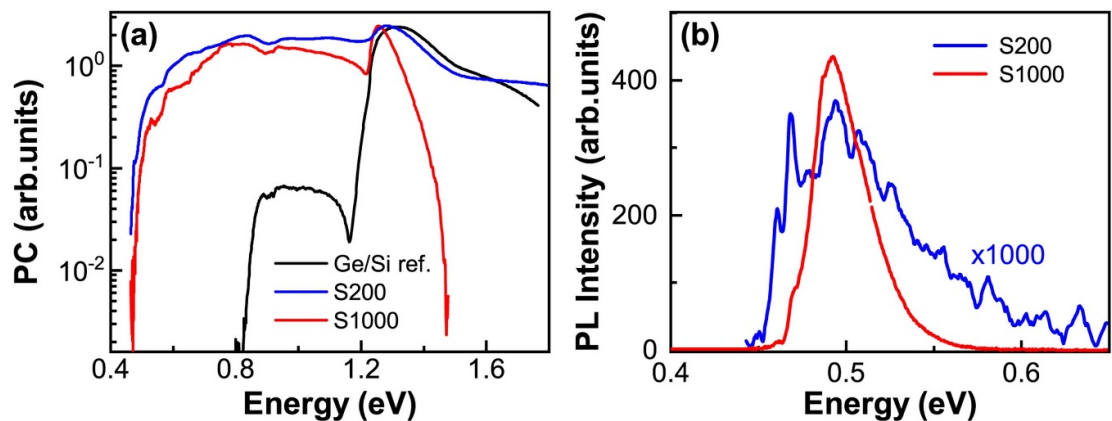


Figure 3. The photoconductivity (a) and photoluminescence (b) spectral dependencies for the studied structures.

bandgap at 80 K. For the other samples, the photocurrent below the Ge absorption edge at 0.73 eV predominantly originates from electron-hole pairs photoexcited band-to-band within the GeSn thin film.

The low-energy onsets of the PC spectra for S200 and S1000 are ~ 0.45 eV, which correlate well with the PL (figure 3(b)). This corresponds to the near-bandgap and bandgap optical transitions in GeSn for the given Sn content [28]. It is worth noting that the PL peak for sample S200 is broader and less intense than the sharper peak observed in sample S1000, with a thicker GeSn film. This generally indicates higher crystal quality of the thicker sample; however, this question requires further attention.

3.3. Impedance Analysis

The electrical characteristics of the GeSn/Ge/Si heterostructures were then investigated through spectroscopy of the complex impedance represented as $Z(f) = Z_1(f) - iZ_2(f)$, where $Z_1(f)$ and $Z_2(f)$ denote the real and imaginary components of the impedance, respectively. This technique is particularly effective for determining the electrical properties of thin films, such as conductivity and carrier relaxation. In the experiment, conductivity was measured by applying an oscillating electric field along the epitaxial layer, thus allowing investigation of current flow in the parallel-connected layers of GeSn, Ge, and

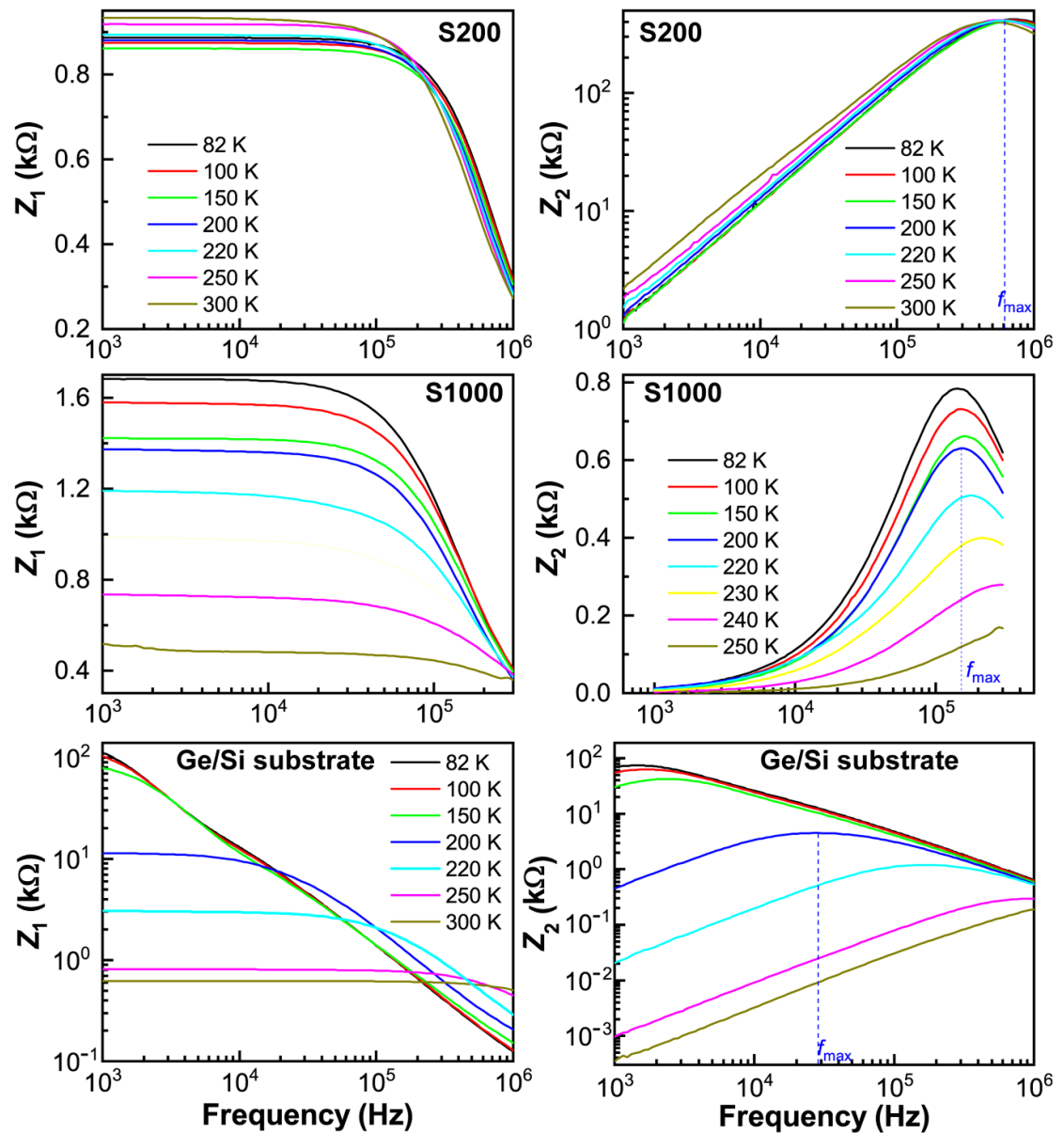


Figure 4. Frequency dependences of the real (left) and imaginary (right) parts of impedance for the studied samples at different temperatures.

Si substrate. This AC voltage was maintained at 10 mV, while the frequency was varied from 1 kHz to 1 MHz for each temperature from 80 to 300 K with steps of 5 K. Figure 4 shows the frequency dependences of the real (Z_1) and imaginary (Z_2) parts of the impedance at different temperatures for both samples S200 and S1000 as well as for the reference Ge/Si structure. Interestingly, at temperatures below approximately 200 K, Z_1 and Z_2 are an order of magnitude lower than those observed for the Ge/Si sample, reflecting the dominant contribution of the top GeSn film to the conductivity. In contrast, the parallel transport in the Ge/Si virtual substrate makes a significantly greater contribution to the complex impedance at higher temperatures.

Z_1 reveals a Sigmoid-like variation with frequency. For the GeSn film (samples S200 and S1000), the values reached a plateau below $\sim 10^4$ Hz, with the magnitude decreasing monotonically with increasing temperature. This represents the temperature-dependent changes in the resistance of the GeSn alloy. Sample S200, however, reveals a slight increase in Z_1 at higher temperatures, and thus, the T -dependence is abnormal for a semiconductor structure, as discussed below. In comparison, the reference Ge/Si structure demonstrates a saturation frequency that correspondingly decreases with temperature, so that no saturation is reached within the measured frequency range at maximum cooling. Meanwhile, the Z_1 values are much higher due to the parallel contributions of the less-conductive Ge buffer and the Si substrate.

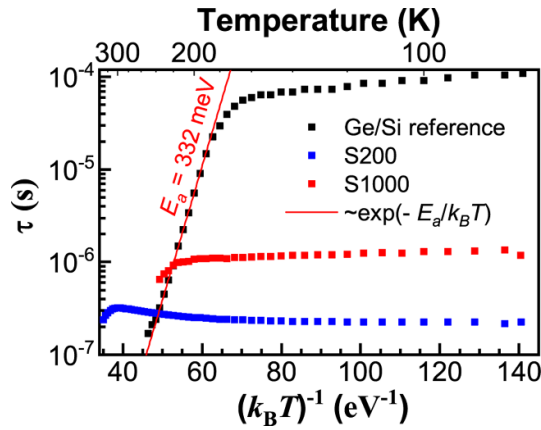


Figure 5. Arrhenius plot of relaxation time (τ) from $-Z_2$ with temperature (T).

The imaginary part of the impedance exhibits a pronounced peak, indicating a resonant frequency, $f_{\max}$, at which energy loss is maximized due to the polarization mechanism. The mean relaxation time was evaluated from this peak position following $\tau = (2\pi f_{\max})^{-1}$ for each temperature [29]. Figure 5 shows the variation of this relaxation time with reciprocal temperature for all samples. The temperature dependence for the buffer-only sample (Ge/Si) in the temperature range 180–300 K follows an exponential law $\tau \sim \exp[E_a \cdot (k_B T)^{-1}]$, where E_a is the activation energy involved in carrier relaxation, and k_B is the Boltzmann constant. The slope of the dependence yields the value of E_a as ~ 330 meV for the reference sample. At lower temperatures, τ changes only slightly and is 2–3 orders of magnitude higher than that in samples S200 and S1000 containing the GeSn films. These two samples also exhibit a similar fall with heating. As a result, we cannot estimate an exact value of E_a from the slope; however, we may assume the same effect as in the reference Ge/Si sample. The value of 330 eV is close to half the Ge bandgap; thus, we consider the high-temperature relaxation of thermally excited carriers due to interband transitions in Ge.

Figure 6 shows the Nyquist plots measured at different temperatures. They are the plots of the imaginary part of the impedance, Z_2 , versus its real part, Z_1 . Each plot is a single semicircle arc, regardless of the temperature at which it was recorded. This means that the electrical processes in the GeSn films can be associated with a simple RC equivalent circuit. The Nyquist plots were fitted using Z-view software to gain a better understanding of the relaxation process. The best fits are obtained using an equivalent circuit involving a parallel combination of resistance and a ‘constant phase element’ (CPE). The relationship gives the impedance of the CPE:

$$Z_{\text{CPE}} = \frac{1}{Q(j\omega)^\alpha}, \quad (2)$$

where Q and α are the parameters of the CPE. As the temperature increases, the semicircle diameter decreases, following the temperature dependencies of the real and imaginary components of the impedance. The resistance, R , at each temperature was estimated from the semicircle diameter (the maximum value of Z_1).

The DC electrical conductance is calculated as the reciprocal of the resistance, $G_{\text{DC}} = 1/R$, and figure 7 shows its temperature dependence. The reference Ge/Si structure exhibits a relatively typical semiconductor response: conductivity increases slowly at low temperatures due to increased carrier mobility. In contrast, the sharp rise for $T \geq 160$ K is related to the thermal activation of free carriers in Ge. Therefore, the temperature dependence of the conductivity for the Ge/Si sample within 160–250 K follows an Arrhenius law:

$$G_{\text{DC}}(T) \propto \exp\left(-\frac{E_a}{k_B T}\right) \quad (3)$$

where E_a is the activation energy of DC conductivity. The calculated energy was ~ 250 meV and can be attributed to carrier emission from deep traps in Ge or the Ge/Si interface. For the GeSn samples, S200 and S1000, the temperature dependence of the conductivity is similar. Moreover, the conductivity at low

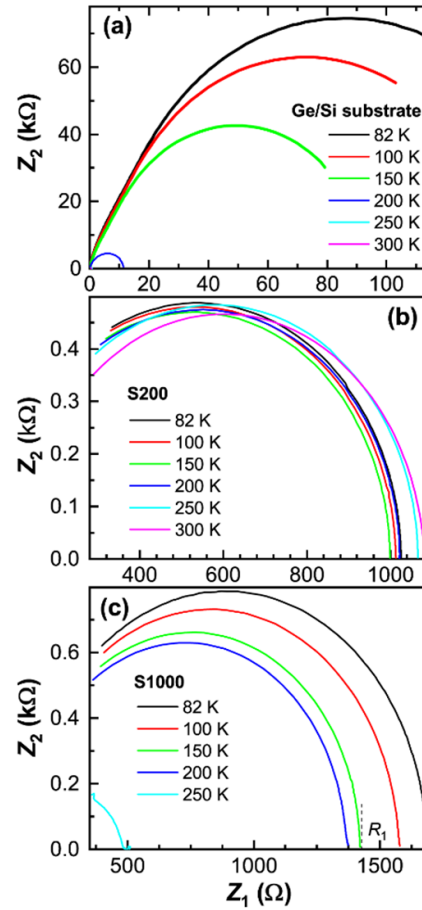


Figure 6. Nyquist plots for the studied Ge/Si reference sample (a), GeSn/Ge/Si structures with GeSn layer of thickness 200 nm (b) and 1000 nm (c) measured at different temperatures.

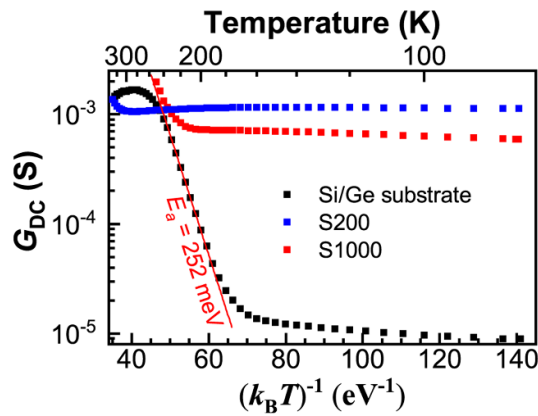
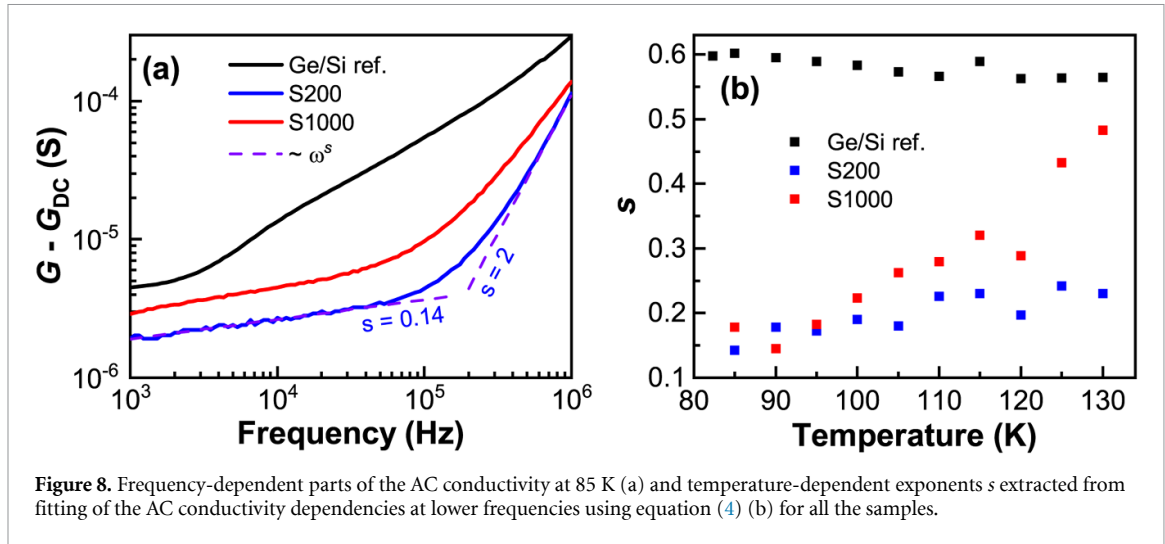


Figure 7. Temperature dependencies of the DC conductance in the studied structures.

temperatures is a few orders of magnitude higher than that of the reference sample, indicating a dominant contribution from the GeSn layer. In turn, high conductivity over a low-temperature range indicates a high density of localized states in GeSn near the Fermi level. These samples exhibit a decrease in DC conductance with increasing T at certain temperatures (more pronounced for S200 within 160–280 K as shown in figure 7), which is abnormal for semiconductors. It should also be noted that the DC conductivity of the 200 nm GeSn layer at low temperatures is higher compared to the more relaxed sample S1000, likely due to a higher dislocation density resulting from strain relaxation.

Considering the much higher conductivity of samples S200 and S1000 compared to the reference structure at low temperatures, it is reasonable to conclude that the conductivity in the Sn-containing



samples is caused mostly by the GeSn film. Taking into account the sample sizes and film thicknesses, we estimated conductivity in the GeSn material in samples S200 and S1000 at liquid nitrogen temperatures as, respectively, ~ 94 and 10 S cm^{-1} , i.e., resistivities of 1.1×10^{-2} and $1.0 \times 10^{-1} \Omega \cdot \text{cm}$, respectively. For comparison, the resistivity of the 700 nm thick Ge buffer at these temperatures was evaluated from the DC conductance of the reference sample to be $\sim 2.4 \Omega \cdot \text{cm}$, assuming the conductivity in the Ge/Si structure at low temperatures to be related to the top Ge layer.

3.4. AC conductivity study

The frequency dependences of the real part of the complex conductivity, $G = Z_1 / (Z_1^2 + Z_2^2)$, for the GeSn/Ge/Si structures at different temperatures were analyzed. At low temperatures, all the frequency-dependent parts of G were found to be sublinear at low frequencies (1–20 kHz), transforming to superlinear at high frequencies (figure 8(a)). Above 10^4 Hz, $G(\omega) \propto \omega^2$ due to the small resistance of the metal contacts [30]. Note that the contribution of the Ge buffer layer to the total AC conductivity increases, and its coupling through the resistance and capacitance of the GeSn/Ge heterojunctions becomes more pronounced, enhancing its overall influence on the AC transport mechanisms at high frequencies. For the lowest temperatures (below 130 K) and frequencies within the 10^3 – 10^4 Hz range, the intrinsic GeSn film response is dominant. Thus, for analyzing the conduction mechanism of the GeSn film, this frequency and temperature range was used for further consideration.

The AC conductivity at low temperatures, and consequently the conductance, has been described by an equation known as the universal dynamic response or Jonscher's power law [31]:

$$G(\omega) = G_{\text{DC}} + A \cdot \omega^s, \quad (4)$$

where A is a temperature-dependent pre-exponential factor, and s is a fractional exponent. The first term corresponds to frequency-independent DC conductivity, which is associated with the conduction path of non-localized charge carriers between electrodes, enabling the current to flow. The second term corresponds to the frequency-dependent AC conductivity, which arises from either hopping across or tunneling through energy barriers between distinct localized states or a combination of both processes [32–34]. In most cases, the fractional exponent typically takes values $0 < s < 1$, while the behavior of $s(T)$ is key in estimating the conduction mechanism. The correlated barrier hopping model, proposed by Elliot [33], predicts that s decreases with an increase in temperature, reflecting reduced effective barrier heights due to thermal activation. Quantum mechanical tunneling (QMT) predicts that the exponent, s , is temperature-independent and can be expressed by

$$s = 1 - \frac{4}{\ln\left(\frac{1}{\omega\tau_0}\right)}, \quad (5)$$

where τ_0 is the characteristic relaxation time.

According to the overlapping large polaron tunneling model [35], the s value decreases to a minimum value when heating, then increases with temperature. The non-overlapping small polaron tunneling (NSPT) model [36] predicts that the exponent s increases with temperature as

$$s = 1 - \frac{4}{\ln\left(\frac{1}{\omega\tau_0}\right) - \frac{W_H}{k_B T}}, \quad (6)$$

where W_H is the polaron hopping energy.

The variation of s at the lower temperatures for the GeSn samples is shown in figure 8(b). The exponent is obtained from fitting by equation (4) of the measured frequency dependencies at various temperatures for all the samples. It was found that the exponent for the reference Ge/Si sample is about 0.6 and changes negligibly, indicating that the QMT model is applicable. At the same time, for the investigated GeSn alloys, the parameter, s , increases from ~ 0.15 to ~ 0.25 and 0.5 for samples S200 and S1000, respectively, with increasing temperature from 85 to 130 K. The slope increases with temperature, which rules out any adequate fitting by equation (6) within the NSPT model, necessitating an alternative mechanism to explain the hopping conductivity. From a general point of view, the tunneling probability is determined by the overlap of wavefunctions between localized states, which depends on the distance between states and the energy difference. The high AC and DC conductivity in the studied GeSn films can be attributed to the transport of charge carriers within a band of localized states that pin the Fermi level, leading to a metallic-type conductivity. The lower values for sample S200 confirm the presence of a hopping process through disorder-induced defect states, reflecting a crossover from band-like transport to a disorder-driven hopping regime with increasing structural disorder. In the absence of intentional doping, the high conductivity at low temperatures can be attributed to a higher concentration of defect-related acceptor states [37], particularly vacancies and vacancy-related complexes formed during strain relaxation near the GeSn/Ge interface. These acceptor-like point defects increase the hole concentration and provide a dense network of localized states that facilitate hopping-dominated charge transport within the low-temperature range, increasing with T . In contrast, the sample S1000, characterized by higher structural quality and a lower density of such defects, exhibits a lower contribution from defect-assisted hopping pathways, resulting in lower DC conductivity.

The tunneling probability is expected to be temperature-independent, as it depends solely on the spatial and energy position of the localized states. The QMT mechanism provides the best agreement with the experimental data in the low-temperature range of ~ 85 – 110 K, where the influence of thermally activated processes is minimal. Some increases in the exponent with temperature may be due to the impact on the overall AC conductivity of other mechanisms, such as VRH and/or phonon-assisted tunneling [38, 39]. These factors increase the effective rate of charge carrier transfer and govern the temperature dependence of the observed AC conductivity as well as a slight increase of s .

In general, the low growth temperatures required for GeSn epitaxy result in the formation of various trap states above the valence band, although their position is rarely assigned unambiguously to specific defects. In addition to the Ge-related defect states commonly observed in Ge and Ge/Si heterostructures, GeSn alloys contain trap states originating from the incorporation of Sn [8, 13]. The coexistence of these deep Sn-related traps and Ge-derived band-tail states enhances carrier localization, promoting hopping-type transport, as observed in our impedance analysis. The low values of s and its non-monotonic temperature dependence observed in the studied GeSn films indicate a high degree of structural disorder and/or the coexistence of parallel transport channels governed by different conduction mechanisms.

3.5. KPFM and SCM measurements

The data obtained indicates the presence of a band of localized states that determines the electrical conductivity of the GeSn film. Considering that the silicon substrate has p-type conductivity and the germanium buffer layer acts as a large-scale trap for holes by forming a potential well in the valence band, an excess of holes is expected in the GeSn film. These holes can fill the band of the localized states, pinning the Fermi level and leading to downward band bending near the surface.

KPFM and SCM measurements were performed to explore this potential and determine the local work function as well as the majority charge carrier type in the surface region of the GeSn film. Figure 9 shows the CPD maps for samples S200 and S1000. The spatial distribution maps of the surface potential revealed slight variations of several tens of meV. The maximum of the CPD distribution histogram for the S1000 sample was observed near 0.29 V and was ~ 0.2 V higher than the position of the histogram maximum for the S200 sample (near 0.10 V). Taking into account that KPFM measurements provide the contact potential difference $V_{CPD} = (\Phi_{tip} - \Phi_s)/-e$, where Φ_{tip} and Φ_s are the electron work functions of the tip and the sample, respectively, and e is the elementary charge, it can be concluded that the work

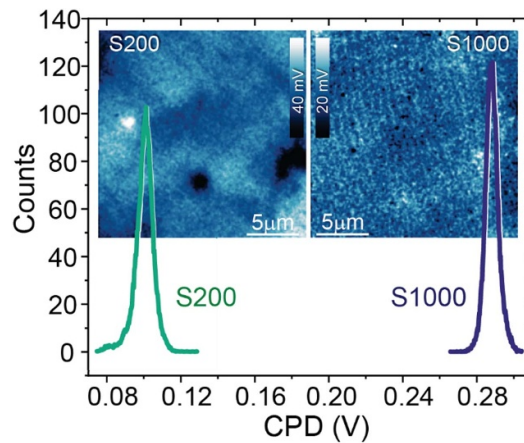


Figure 9. Histograms of KPFM-measured contact potential difference (CPD) for samples S200 and S1000. Insets display the corresponding CPD maps.

function of the S1000 sample is 0.18 eV lower. This reduction is probably enough for the transition to n-type conductivity in the GeSn alloys with a bandgap of 0.46 eV (determined from PC and PL spectra).

The SCM technique was then used to visualize the spatial distribution of the active charge carriers by measuring dC/dV maps under an applied AC voltage between the probe tip and the GeSn film. Figure 10 shows the SCM images of GeSn films illustrating the opposite polarity of dC/dV amplitude and, therefore, an opposite type of majority charge carriers in the studied samples. In particular, the dC/dV signal was found to be positive for sample S200, indicating p-type conductivity. Since dC/dV amplitude is inversely proportional to the carrier concentration of the GeSn layer, its behavior indicates the inhomogeneous distribution of free charge carrier concentration. In contrast, the map of sample S1000 shows mainly regions with n-type conductivity, with negative dC/dV amplitudes; however, regions with p-type conductivity are also observed. The findings indicate that the Fermi level lies near the middle of the bandgap, closer to the conduction band, while spatial variations of electrostatic potential are responsible for the coexistence of both n- and p-type regions. This phenomenon likely explains the lower electrical conductivity observed in this sample compared to S200, which is identified as more defective, where holes dominate as the majority carriers uniformly across all surface regions. SCM images for GeSn films resemble the cross-hatch patterns observed in AFM images. These patterns are observed along the perpendicular [110] and $[-110]$ directions and originate from the periodic strain fields induced by the influence of misfit dislocations at the GeSn/Ge and/or Ge/Si interface [18, 28]. Along with other defects near the GeSn/Ge interface, this factor likely contributes to the p-type conduction and the 0.18 eV lower position of the Fermi level observed in the thinner sample S200.

The dC/dV phase shift is further analyzed to distinguish the polarity of the majority carrier. The histograms shown in figure 10(e) and figures S1–S4 in the supporting information reveal significant free carrier concentrations. The positive values of the detected dC/dV phase confirm the dominance of p-type conductivity in sample S200. The histogram maximum for the dC/dV phase shift was observed near 80 deg., indicating that the hole concentration is much higher than for sample S1000, which has higher crystal quality and less dislocation evidence at the AFM topology.

The higher concentration of holes and, consequently, the higher AC and DC conductivity of sample S200 are attributed to its lower thickness and, hence, more influence of the interface with Ge with a high density of acceptor-type defects associated with misfit dislocations. Interface states can participate in hopping conduction, whereas variations in the electrostatic potential shift their positions relative to the Fermi level, forming a band of states that provide more possibilities for AC transport. The low values of s in the universal dynamic response further confirm the presence of structural imperfections. The observed increase in s with heating is attributed to the filling of the defect-related band with electrons, which suppresses the contribution of interface defect states to the hopping process.

The GeSn films contain misfit dislocations mainly at the interface region that extends into the bulk, forming 90° Lomer and 60° mixed dislocations. Notably, the layer near the GeSn/Ge interface is the most defective, indicating that the high-quality GeSn layer can be grown far from the Ge buffer layers [12]. It would be speculative to assume that the GeSn/Ge interface in the thicker sample S1000 differs from that in S200 and contains other amounts and/or types of defects. Hence, the thicker film with such a conductivity channel should exhibit higher or at least similar conductance; however, we observed an

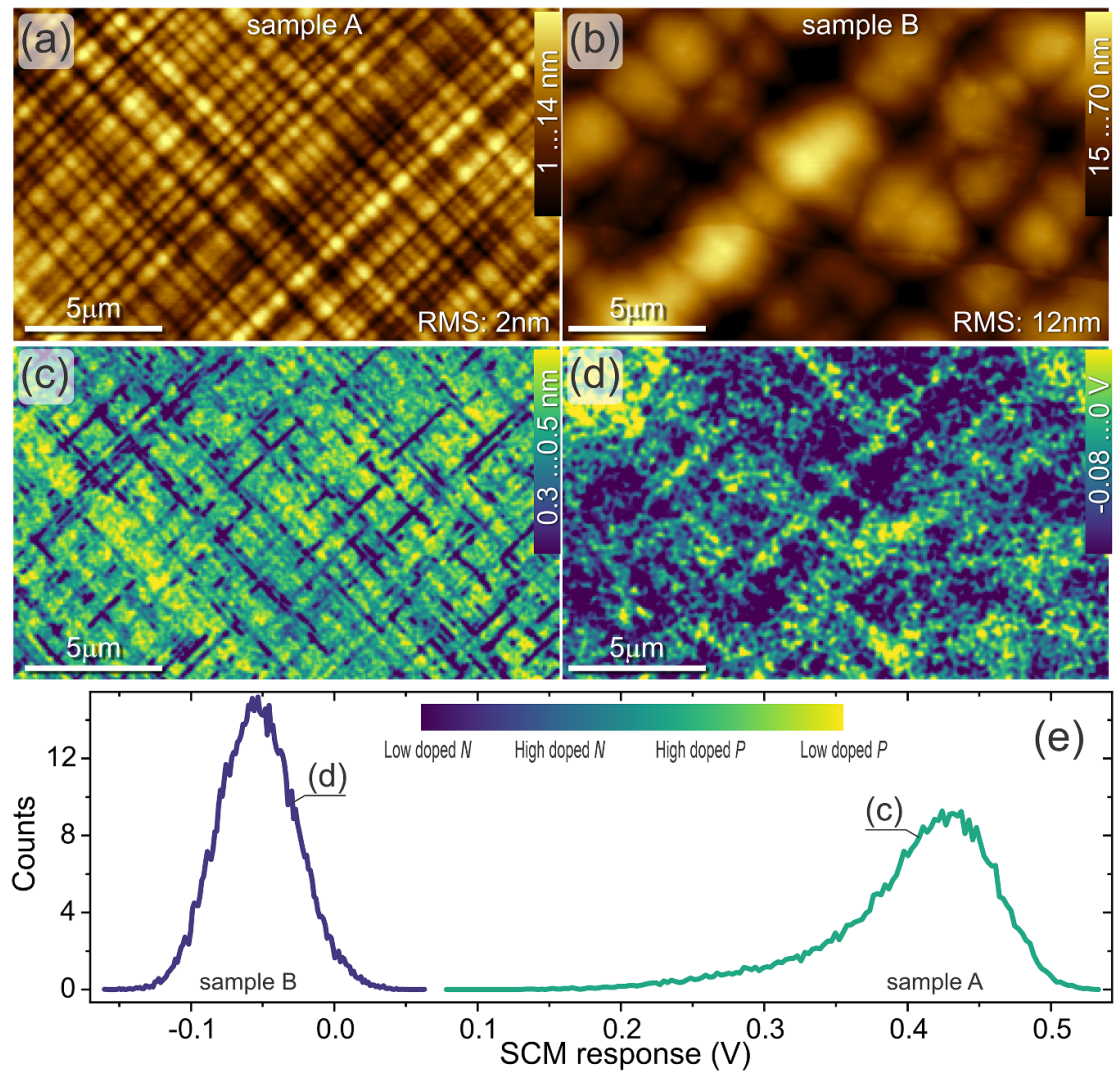


Figure 10. AFM (a), (b) and SCM (c), (d) images for GeSn films with a thickness of 200 nm (a), (c) and 1000 nm (b), (d). Histograms of the dC/dV phase shift are presented in (e).

order of magnitude lower DC conductivity at low temperatures. Obviously, the hopping channel in the sample with the thicker GeSn film is inhibited.

We assume that the GeSn films contain a number of donor-like defects that can affect the charge carrier statistics. Their influence in thin films is not so pronounced, and the film retains p-type conductivity. However, the amount of these defects in the thicker film is higher, and the Fermi level rises. This leads to an increased filling of the defect band states with holes, reducing the number of hopping pathways, particularly near the Ge/Si interface. As a result, the lower conductivity is observed in sample S1000, which exhibits the coexistence of spatially inhomogeneous n- and p-type regions. The presence of such mixed carrier-type regions can significantly deteriorate macroscopic device performance, leading to the formation of local potential barriers or depletion regions that restrict carrier transport. These electrically inhomogeneous regions may be a reason for non-ideal impedance behavior, with an exponent s much less than 1 in Jonscher's universal power law.

In this work, we combine temperature- and frequency-dependent AC conductivity measurements with nanoscale KPFM and SCM mapping to establish a correlation between complex conductivity and the spatial distribution of localized states in GeSn films. The AC conductivity analysis provides insight into the role of disorder-induced hopping pathways, while KPFM and SCM reveal nanoscale variations in surface potential and carrier concentration that originate from structural disorder and defect-induced electrostatic fluctuations. By integrating these complementary techniques, we describe frequency-dependent transport and microscopic inhomogeneities, thereby providing insight into how local variations in electronic structure influence the macroscopic electrical response of GeSn alloys.

4. Conclusions

In the present study, we investigated the complex AC conductivity of GeSn alloys within the frequency range 1 kHz–1 MHz and temperature range 85–330 K. Impedance spectroscopy, combined with structural, optical, and scanning probe techniques, reveals that electrical transport in GeSn alloys at low temperatures is dominated by hopping conduction via localized states. The thicker (1000 nm) GeSn film was found to have a more perfect crystal structure and higher Fermi level position than the thinner (200 nm) one, while revealing an order higher resistivity of $1.0 \times 10^{-1} \Omega\cdot\text{cm}$ below ~ 200 K, which is, however, more than order lower than that in the underlying Ge buffer estimated to be above $2 \Omega\cdot\text{cm}$. This behavior suggests that charge transport in GeSn films is significantly influenced by the presence and distribution of defect-related localized states, which facilitate hopping conduction.

The AC conductance behavior of GeSn alloys at low temperatures is similar to that of disordered materials, revealing a dominant contribution from hopping conduction. The combination of KPFM and SCM measurements reveals the spatial variations in carrier concentration that originates from the presence of disorder-induced localized states. The frequency dependence of the AC conductivity follows Jonscher's universal power law, with an exponent less than 1 in the frequency range of 1–20 kHz, which is consistent with hopping conduction through a defect-related band. These states pin the Fermi level and significantly influence AC transport, particularly in more defective films. The results highlight the critical role of structural disorder and defect distribution in determining the electrical behavior of GeSn films, providing valuable insights for optimizing these materials in optoelectronic applications.

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Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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Author contributions

Serhiy Kondratenko  0000-0002-4403-7732

Conceptualization (equal), Methodology (equal), Writing - original draft (equal)

Oleksandr Datsenko  0000-0002-4461-1925

Formal analysis (equal), Investigation (equal), Visualization (equal), Writing – review & editing (equal)

Fernando M de Oliveira

Formal analysis (equal), Investigation (equal), Visualization (equal)

Serhii Kryvyi

Formal analysis (equal), Investigation (equal)

Kateryna Yablochkova  0000-0001-8719-5776

Investigation (equal), Methodology (equal), Writing – original draft (equal)

Petro M Lytvyn

Conceptualization (equal), Funding acquisition (equal), Investigation (equal), Writing – review & editing (equal)

Morgan E Ware

Formal analysis (equal), Investigation (equal), Writing – review & editing (equal)

Serhii Maliuta  0000-0003-0466-9900

Formal analysis (equal), Investigation (equal), Visualization (equal)

Shui-Qing (Fisher) Yu

Formal analysis (equal), Methodology (equal)

Yuriy I Mazur

Formal analysis (equal), Methodology (equal), Writing – review & editing (equal)

Gregory J Salamo

Conceptualization (equal), Methodology (equal), Writing – review & editing (equal)

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